

Miranda Gemme-Ellis

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Willing to relocate

EDUCATION

Merrimack College - North Andover, MA

October 2024

Master of Science in Data Science

GPA: 3.6

Relevant Courses: R and Python Programming, Predictive Modeling, Visual Data Exploration, Machine Learning and Artificial Intelligence, Social Media and Marketing Analytics, Data Governance and Privacy, Data Science Capstone

University of Massachusetts - Amherst, MA

May 2020

Bachelor of Science in Public Health Sciences

Concentration: Epidemiology

Relevant Courses: Biostatistics, Data Collection & Analysis, Telling Stories with Data: Modeling and Visualization, Epidemiology, Chronic Disease Epidemiology

Holyoke Community College - Holyoke, MA

June 2018

Associate of Arts in Liberal Arts and Sciences

Relevant Courses: Statistics, Algebra, Calculus, Psychology

SKILLS

Data Science & Analytics: Data Analysis, Data Visualization, Data Mining, Statistical Modeling, Machine Learning, Quantitative & Qualitative Research, Survey Design, Data Wrangling, Data Cleaning

Technical Skills: Python (NumPy, pandas), R, SQL, Jupyter Notebook, Tableau, Stata, GitHub, Salesforce, Qualtrics, Nikon Elements, Qualtrics (survey development and analysis), Salesforce (CRM and analytics integration), Tableau (dashboards and visual reporting), GitHub (version control and collaboration)

Core Competencies: Analytical Thinking, Quantitative Reasoning, Attention to Detail, Problem Solving, Critical Thinking, Written and Verbal Communication, Data Interpretation, Stakeholder Collaboration, Time Management, Adaptability, Decision-Making, Technical Writing

DATA SCIENCE & MACHINE LEARNING PROJECTS

Sentiment Analysis Using Python

October 2024

Social Media and Marketing Analytics

- Created and analyzed a social network dataset using Python's NetworkX library to calculate key graph metrics, including Degree, Betweenness, Closeness, and Eigenvector Centrality, identifying the most influential nodes in the network.
- Explained the significance of centrality measures in social network analysis, demonstrating insights into relationship patterns and node influence within a connected graph.

Social Determinants of Health Impact on Quality of Life in US Counties

July 2024-August 2024

Data Science Master's Program Capstone

- Worked in a team of 3 graduate students on a predictive analysis project analyzing how Social Determinants of Health (SDOH) influence the quality of life across U.S. counties.
- Performed comprehensive data acquisition and cleaning using R by merging two large datasets from the Agency for Healthcare Research and Quality (AHRQ) and County Health Rankings (CHR) to create a dataset of over 3,000 counties.
- Built and tested several machine learning models, identifying the Gradient Boosting Machine (GBM) as the most accurate predictor.
- Documented code, analysis, and deliverables in GitHub to ensure reproducibility and version control.
- Presented key findings and actionable recommendations to stakeholders in a 34-page formal report and presentation to guide public health policy interventions.

Hotel Booking Cancellation Prediction Using Machine Learning

June 2024

Machine Learning and Artificial Intelligence

- Built predictive models using logistic regression and neural networks to analyze 35,000+ hotel booking records, achieving an AUC of 0.87 for the logistic regression model and 0.78 for the neural network model.
- Developed and fine-tuned neural network models using regularization techniques, including dropout layers and L2 regularization, to improve generalization and reduce overfitting.

- Utilized Python, R, and TensorFlow for data preprocessing, model development, and evaluation.

Predicting Medical Expenses and Smoking Likelihood with Health and Demographic Data

December 2023

Predictive Modeling

- Analyzed a health insurance dataset with 1,300+ records, applying Ridge, Lasso, and Gradient Boosting models to predict medical expenses and smoking likelihood. Achieved a reduction in prediction error by 12% using Gradient Boosting.
- Preprocessed and engineered data in R, addressing missing values, transforming variables, and encoding categorical features to improve model performance.
- Delivered actionable insights for healthcare providers, proposing strategies to reduce costs by promoting healthy BMI, implementing smoking cessation programs, and addressing regional healthcare disparities.

WORK EXPERIENCE

CVS Pharmacy - Granby, MA

June 2023-January 2025

Lead Pharmacy Technician

- Managed pharmacy operations and ensured compliance with federal regulations.
- Handled patient data securely, adhering to HIPAA guidelines, and optimized workflows to improve service delivery.

Pharmacy Technician

December 2021-June 2023

- Managed inventory by verifying quantities, removing outdated medications, and restocking pharmaceuticals to prevent being out of stock.
- Collaborated with pharmacists to calculate accurate prescription dosages, dispense medications, and fill prescriptions.

University of Massachusetts Donahue - Hadley, MA

November 2019-October 2021

Student Research Assistant

- Assisted with data collection through surveys, interviews, and focus groups to support research initiatives.
- Performed data entry, cleaning, and quality assurance to ensure data integrity and reliability.
- Analyzed quantitative and qualitative data, contributing insights and findings to research projects.
- Created detailed charts and tables to effectively visualize research results and trends.

Partners In Health - Boston, MA

December 2020-March 2021

Case Investigator

- Analyzed patient and business data to investigate COVID-19 clusters and identify epidemiological links.
- Collaborated with outreach specialists to contact community members and businesses for COVID-19 spread mitigation.

Contact Tracer

November 2020-December 2020

- Conducted interviews with COVID-19 patients and their contacts to collect epidemiological data using Salesforce.
- Informed patients and contacts about testing sites, quarantine procedures, and daily symptom monitoring.

Randall's Farm - Ludlow, MA

April 2014-November 2020

Greenhouse Worker

- Provided exceptional customer service by offering plant and fertilizer expertise to customers.
- Trained and supervised new employees on daily opening and closing procedures to ensure smooth operations.

University of Massachusetts - Amherst, MA

June 2019-March 2020

Light Microscopy Technician Intern

- Collaborated with interns to test Nuclear Staining Dyes on mammalian cells for research.
- Updated software and backed up research data to maintain computer systems efficiently.
- Revised programming for HCA Nikon TiE microscope to enhance data collection and analysis.